

-- 1. CREATE: Create database and table

CREATE DATABASE company;

USE company;

CREATE TABLE employees (

emp_id INT PRIMARY KEY AUTO_INCREMENT,

emp_name VARCHAR(50) NOT NULL,

department VARCHAR(30),

salary DECIMAL(10,2),

hire_date DATE

);

```
mysql> CREATE TABLE employees (
->     emp_id INT PRIMARY KEY AUTO_INCREMENT,
->     emp_name VARCHAR(50) NOT NULL,
->     department VARCHAR(30),
->     salary DECIMAL(10,2),
->     hire_date DATE
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> desc employees;
```

Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	auto_increment
emp_name	varchar(50)	NO		NULL	
department	varchar(30)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
hire_date	date	YES		NULL	

5 rows in set (0.03 sec)

-- 2. Altering a table

Modify FACULTY by adding column Dept VARCHAR(10)

ALTER TABLE FACULTY ADD Dept VARCHAR(10);

ALTER TABLE FACULTY ADD Age NUMBER(3);

-- 3. ALTER TABLE - DROP COLUMN (demonstration) -----

ALTER TABLE STUDENT ADD TempCol VARCHAR(5); -- add a throwaway column

ALTER TABLE STUDENT DROP COLUMN TempCol; -- then drop it

```
mysql> ALTER TABLE FACULTY ADD Dept VARCHAR(10);
Query OK, 0 rows affected (0.10 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE FACULTY ADD Age NUMBER(3);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
gives you the right syntax to use near 'NUMBER(3)' at line 1
mysql> ALTER TABLE STUDENT ADD TempCol VARCHAR(5);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE STUDENT DROP COLUMN TempCol;
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | YES | | NULL | |
| FacName | varchar(15) | YES | | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| DOJ | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
| Dept | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.01 sec)
```

4. Adding Constraint

ALTER TABLE STUDENT ADD PRIMARY KEY (RegNo);

ALTER TABLE FACULTY ADD PRIMARY KEY (FacNo);

ALTER TABLE DEPARTMENT ADD PRIMARY KEY (DeptNo);

ALTER TABLE COURSE ADD PRIMARY KEY (CourseNo);

```
mysql> ALTER TABLE STUDENT ADD PRIMARY KEY (RegNo);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE FACULTY ADD PRIMARY KEY (FacNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE DEPARTMENT ADD PRIMARY KEY (DeptNo);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE COURSE ADD PRIMARY KEY (CourseNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo | varchar(3) | NO | PRI | NULL | |
| CourseDesc | varchar(14) | YES | | NULL | |
| CourseType | char(1) | YES | | NULL | |
| SemNo | char(1) | YES | | NULL | |
| HallNo | varchar(4) | YES | | NULL | |
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
```

-- 5. ALTER TABLE - ADD FOREIGN KEY constraint -----

ALTER TABLE COURSE ADD CONSTRAINT fk_course_faculty

FOREIGN KEY (FacNo) REFERENCES FACULTY(FacNo);

ALTER TABLE DEPARTMENT ADD CONSTRAINT fk_dept_head

FOREIGN KEY (DeptHead) REFERENCES FACULTY(FacNo);

```
mysql> ALTER TABLE COURSE ADD CONSTRAINT fk_course_faculty
-> FOREIGN KEY (FacNo) REFERENCES FACULTY(FacNo);
Query OK, 0 rows affected (0.12 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql>
mysql> ALTER TABLE DEPARTMENT ADD CONSTRAINT fk_dept_head
-> FOREIGN KEY (DeptHead) REFERENCES FACULTY(FacNo);
Query OK, 0 rows affected (0.10 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

-- 6. RENAME TABLE -----

RENAME TABLE DEPARTMENT TO DEPT;

```
mysql> RENAME TABLE DEPARTMENT TO DEPT;
Query OK, 0 rows affected (0.03 sec)

mysql> desc dept;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DeptNo     | varchar(4)    | NO   | PRI | NULL    |       |
| DeptName   | varchar(15)   | YES  |     | NULL    |       |
| DeptHead   | varchar(4)    | YES  | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.06 sec)
```

-- 7. DROP TABLE -----

DROP TABLE SCRATCH;

```
mysql> CREATE TABLE SCRATCH (id INT);
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO SCRATCH VALUES (1), (2), (3);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> TRUNCATE TABLE SCRATCH;
Query OK, 0 rows affected (0.04 sec)

mysql> drop table scratch;
Query OK, 0 rows affected (0.04 sec)
```